# Audit and Review Record: Evidence Versions and Execution History

Public review revision, 14 October 2026; authorship unconfirmed

## 1 13 October corrections and evidence boundaries

The new reader manuscript addresses the status, all-50-table and six-map reviews. The 9 October public PDFs, including the 77-page Chinese supplement, remain unchanged in `revision46-review-v1`. Corrected PDFs are separately versioned; the historical review checklist is not rewritten as if it had always contained the corrected statements.

The previous main-text ES denominator $2N^2$ and the corresponding checklist sentence were erroneous. Execution content beginning `c3e44393`, the retained runtime settings and the public paired-replay receipt bind `metrics.py` to SHA-256 beginning `c00059e4`; the common scorer `formal_scoring.py` matches its execution hash beginning `d39adc6d`. Its call chain uses the ordered-pair U statistic with $2N(N-1)$, as already printed in the scientific supplement. The point-mass inertial scorer has a separate direct-distance branch. Full hashes and the two pure source definitions are in `reader-v2-score-binding.json`. This is a source-binding and manuscript correction, not a new raw-array audit or a recalculation of scientific scores. A synthetic two-particle regression distinguishes this operator from the empirical-distribution estimator.

G01 now uses the actual built/vegetation/water–wetland/valid-other groups and no longer claims to depict slope magnitude. The main body adds input geometry, the ten-configuration design, all 21 score/error contrasts, calibration/size and illustrative seed-versus-block uncertainty. Heatmaps retain the same values and role ordering, with pale colours, black text and a symmetric signed coverage-deviation scale. Table captions now distinguish historical pending states from completed scalar/audit receipts; missing second-stream bootstrap details are not reconstructed.

R05 (proxy and label mismatch), R06 (penalty/representation confounding), and R10 (missing task-matched probabilistic baseline) explicitly remain scientific gaps in `revision46-response-v3.json`, linked to the already proposed N03, N04 and N01 designs. No such experiment is executed here. The six private maps retain the original cases, times, 512 particles per panel and failure examples, while the local layout gives shared legends, readable labels and differentiated route/disk lines. Maps, coordinates and private manifests remain outside this public package. True authorship, external-reader feedback, route permissions and human acceptance remain unconfirmed.

## 2 Current reader names and original saved identifiers

Current main-paper and scientific-supplement figures/tables use one naming policy. The historical identifiers below are preserved only for provenance; they are not additional algorithms. In particular, HSM is heading-stratified mixture SDE and RDM is residual-direction mixture SDE. Both use the shared SDE structure, with the fitting and component-schedule differences specified in the main paper. Numbered references retain their original comparison roles.

| Current reader name | Original saved name | Meaning |
|---|---|---|
| HSM-01 | Full01; arm-01/full | Main ordinary reference |
| HSM-07, HSM-11 | Full07, Full11 | Score and training references |
| HSM-16, HSM-18, HSM-20 | Full16, Full18, Full20 | Extra/numerical references |
| RDM | GMM; arm-04/gmm_kernel | Residual grouping and interval redraw |
| CV | Inertial/I | Deterministic constant velocity |
| Heading/redraw | pointwise / pointwise_mixture | Heading groups; interval redraw |
| Speed/radial | explicit_decomp | Explicit speed/radial features |
| Obs $\tau$ s | dt30/dt60/dt120/dt300/dt600 | Observation interval, not integration step |
| Euler-M.; Euler alias | em; euler | Numerical integrator and its alias |

For terrain results, B denotes base, A denotes all-terrain, A minus a named group denotes its loo-group key, and B plus that group denotes its lio-group key. Cover denotes the worldcover key; surface denotes selected surface geometry, not absolute elevation or all terrain. Saved JSON/CSV keys and archived PDFs/manifests remain unchanged. The current rendering entry point is `scripts/render_pirc17_reader_v9.py`; the display-name policy is `scripts/pirc17_display_names.py`. Old plotting scripts are retained as versioned provenance, not the current reader-facing rendering command.

## 3 How to read the record

This record separates scientific arithmetic, execution completion, document editing and human acceptance. The 9 October 2026 revision preserves original results, protocols and all five retained failures. Source unit migration is recorded in `revision46-source-migration.json`. Verbatim English and Chinese pre-revision sources remain `historical-main-v1.tex`. Historical pending counts and old declaration flags are dated snapshots, not today's run queue. No failed forecast was rerun, no successful audit was repeated, and no new fit, forecast, seed or bootstrap budget was introduced.

There are 46 unique recording-hash blocks and 58 reused origin–mode cases, not 58 independent records. Historical fields such as "closed_blocks" and "support_blocks" name program groups; their original JSON meaning is not silently upgraded to participant identity. The 11,368 scoring slots include 11,363 successes and five failures. Required failures invalidate four family/mode cells, including both primary terrain families. Individual failure roots are unknown where stacks or resource/exit records are missing; controller summaries do not prove numerical divergence, OOM, an exact timeout or a physical outage. Private paths and raw routes are not included in this public record.

## 4 Reproduction levels and preserved decisions

Document compilation, saved-aggregate rendering, fixed-stream statistical replay and permission-controlled saved-output audit are separate levels. The original independent audit and subsequent full public fixed-stream replay have readable reports and bindings in `qualification-v1`, `public-paired-replay-v1` and `final-results-v1`. The original 42 review IDs and criteria remain in `review-response-v1.json`; `review-current-v2.md` preserves the subsequent author classifications, not reviewer acceptance. This revision's 46-item record must distinguish edits, automated

checks, scientific gaps and genuine permission/acceptance decisions. A completed editing item is not a solved scientific limitation or an accepted paper.

## 5 Publication choices requiring real information

No author, affiliation, contribution statement, conflict declaration or ethics identifier is inferred. Route publication and each map layer's redistribution scope require distinct real permissions. Until confirmed, only already-public aggregates and anonymous replay records belong in the PR. Local real-map review material must not be uploaded automatically. The owner's requested paper revision authorises PR delivery, not fictional reviewer approval or scientific reruns.

## 6 Retained version, cost and execution material

The following units preserve the detailed old accounts. Current closed status is identified in their audited update; older queued counts describe their own historical captures only.

## 7 Terrain ablation results

The registered complete-family disposition is unavailable, not waiting for remaining forecasts: retained required failures prevent overall/LOO and LIO inference under the unchanged rule. This is not a zero-effect or equivalence result. The overall comparison links to `terrain:all-vs-base`. Four group summaries link to `terrain:road`, `terrain:river`, `terrain:worldcover` and `terrain:surface`. Their evidence covers the primary full-model removal, its baseline addition, input dimensions, coverage and missingness, paired effect and uncertainty, stability, evidence qualification, verdict, cost and limitation.

The all-terrain comparison answers whether the registered combination improves on the history baseline. Leave-one-out comparisons answer whether each group adds incremental predictive information in that combination. Leave-one-in comparisons answer whether a group informs the baseline on its own. These answers will be reported together, without selecting a more favourable comparison after observing results. Six-block secondary summaries will describe origin-mode sensitivity without assigning secondary factor verdicts.

The closed score inventory in Table 8 contains every planned terrain slot. For the 46 primary recording-hash blocks and five fixed forecast seeds, the six configurations required by the overall/leave-one-out family contribute 1380 rows: 1379 successful, one failed and none missing. The five configurations in the supporting leave-one-in family contribute 1150 rows: 1148 successful, two failed and none missing. These families share all 230 baseline rows; their union is 2300 distinct primary terrain forecasts, with 2297 successes and three retained failures, not 2530 independent forecasts or observations. The count-only projection is `terrain-family-completeness-v1.json`; it uses the same closed score index and original comparison membership. The earlier 160/131 missing-row counts in Section 15 are retained as a dated snapshot, not the current unfinished forecast count.

Despite terminal execution, both entire primary-mode terrain families remain unavailable under the registered complete-pair rule: the all-terrain failure affects the overall and all four leave-one-out contrasts; the leave-one-in failures affect the supporting family. No terrain effect or interval is estimated from their successful intersections. Consequently, these results do not establish that roads, rivers, land cover or surface features improve, harm or are equivalent to the history baseline. The completed auxiliary trials and saved-output audit do not replace failed forecasts or restore

these comparisons. The closed counts are a saved-status description, not an independent raw-output audit or final evidence-card acceptance.

**Terminal execution and available inference.** Under the current no-automatic-retry policy, all original scientific forecasts are terminal and recorded failures are retained. A registered comparison family containing any required failed row remains unavailable even when every planned task has a terminal disposition. Completion of execution and availability of paired inference are therefore different conditions. We do not fill failures with zeros, delete failed rows or redefine a registered family to manufacture a result. An unavailable comparison is neither a zero effect nor evidence of equivalence.

**What the five failures actually establish.** Table 1 identifies the retained configurations and forecast seeds. The original checkpoint contains three generic controller interruptions and two in-flight reservations without a completion reply. Neither checked checkpoint root contains an item reply for these five jobs. This is an observation of those two roots, not a claim that every possible historical log has been searched. The last serial request identifies `lio-river`; its closure receipt records zero active owned processes. The two later jobs are bound to distinct predecessor workers and retained 180 s reservations. Reserved seconds are conservative accounting entries, not measured prediction times or evidence of a 180 s timeout.

Table 1: Five retained execution interruptions. Labels are local to this description; no failed output is replaced by a successful subset.

| Label | Configuration | Origin mode | Forecast seed |
|---|---|---|---|
| F01 | `arm-06/dt30` | `known_velocity` | 20260814 |
| F02 | `arm-03/single_gaussian` | `known_velocity` | 20260817 |
| F03 | `lio-river` | `causal_prefix` | 20260817 |
| F04 | `all-terrain` | `causal_prefix` | 20260815 |
| F05 | `lio-road` | `causal_prefix` | 20260814 |

During the method phase, a contemporaneous run summary records a Windows failure to replace a shared request file; it does not supply a separate exception trace for each of F01 and F02. Another summary associates the river interruption with the old supervisor's machine-wide available-memory threshold. The latter summary explicitly distinguishes the threshold stop from an actual allocation failure. These are run-summary observations, not recovered item-level exception traces; the generic saved reason alone cannot decide between interruption, deadline and error. The two unacknowledged worker jobs were discovered after the reported host power loss, but their receipts do not independently diagnose the physical outage. None of these records establishes numerical divergence, a specific invalid simulated path or an intrinsic model failure rate. The saved wording and costs remain unchanged. `paper/pirc17/failure-description-v1.json` binds the failure-map snapshot, predecessor metadata, supporting summaries and frozen comparison sources; this description opens no forecast arrays and performs no retry or output audit.

This is a family-level disposition, not merely the removal of one contrast: `all-terrain` belongs to every primary terrain comparison, and the LIO family also requires its full registered grid. Consequently, the current run cannot support the registered overall-terrain or individual-factor verdicts. Even unfailed LIO pairs are not silently promoted to a new family. The five primary-mode method families contain none of these failures; their already complete 6440-row grid remains eligible for the planned independent replay and qualification, not automatically a final accepted claim.

Table 2: Consequences of the five retained failures under the frozen complete-family rule, even after all runnable jobs finish.

| Registered family | Origin mode | Failed labels | Unavailable contrasts |
|---|---|---|---|
| Terrain primary/LOO | `causal_prefix` | F04 | All five |
| Terrain supporting LIO | `causal_prefix` | F03, F05 | All four |
| Method structure | `known_velocity` | F02 | All four |
| Method interval | `known_velocity` | F01 | All four |

Other secondary families are not invalidated by these five jobs, but retain their own completeness and descriptive-only conditions. Real-map terrain examples remain illustrations, not substitutes for the unavailable population comparisons. We report this evidence limit without adding seeds, refitting, retrying failures or claiming terrain benefit.

# 8 Computational cost

Training, scientific rollout, common scoring, saved-output analysis and paper rendering are measured as separate activities. The two concurrent prediction workers share fitted parameters but maintain separate map-query and history state. Summed worker elapsed cost differs from concurrent wall throughput. Execution-record and reconstruction details are collected in Appendix 17.

# 9 Recorded environment and ordinary prediction-time examples

The recorded CPU environment has Windows 10, six physical and 12 logical CPUs, 31.73 GiB RAM, Python 3.10.13, NumPy 1.26.4 and PyTorch 2.3.1+cu121. The CUDA-enabled package name does not mean these CPU forecasts used a GPU. Library threads are distinct from the two forecast workers; the complete environment and thread settings are given in Appendix 16.

Table 3: Original ordinary prediction-time records at the first registered primary origin and first RNG seed. One recorded run per row, not p50/p95 or an isolated latency comparison; no extra prediction was performed. Seconds are rounded to three decimals for display only.

| Configuration | Recorded prediction seconds |
|---|---:|
| Full01 | 0.973 |
| GMM kernel | 0.817 |
| dt300 | 0.862 |
| EM | 0.676 |
| MC | 0.572 |

Each predicts the nominal 30-minute task with 512 paths and maximum 5-second steps. The original `prediction_seconds` field measures prediction execution with inputs and models already prepared, not all recovery, fitting, disk output, scoring or paper work. These five ordinary records demonstrate that timing exists; a single run does not establish speedup, tail latency or deployment cost. Full recorded precision is retained in `paper/pirc17/preliminary-reader-presentation-v1.json`; it is not a claim of microsecond measurement accuracy. The 75 cold and 75 warm trials and their 15 excluded warmups have all completed and appear in the original review package; they

are not replaced by this example table. Recorded elapsed summaries do not certify uninterrupted isolated latency.

## 10 Execution environment and timing boundary

Timed trials start with prepared causal inputs and fitted parameters already resident. They include provider initialization, prediction setup, the full rollout and prediction-time map reads and queries. Original input and checkpoint preparation, fitting, offline scoring, artifact export and validation, and provider teardown are outside this latency measurement; their costs are not zero. Prediction latency, summed computation and concurrent wall-clock completion time therefore answer different questions.

The isolated runtime experiment uses all ten terrain configurations and five registered method subjects at the first primary origin and seed. Each condition has five provider-cold trials, one excluded warmup and five warm trials. For terrain, provider-cold creates a new map provider; warm reuses the same provider after the subject's warmup. Method operations use resident parameters without a mutable map provider. The Python process remains running, and the operating-system cache is neither reset nor verified; provider-cold is not an OS-cold or cold end-to-end deployment measurement.

The recorded p50 and p95 are total successful-trial latency in milliseconds. A separate mean is normalized by the actual final target time $H$, in seconds: $\overline{C}/(H/60)$ milliseconds per forecast minute, where $\overline{C}$ is the mean successful-trial latency in milliseconds. This normalization describes computational cost, not accuracy or physical clock certification. Failures remain in the five-trial failure-rate denominator; warmup is excluded from latency quantiles. Five trials yield descriptive quantiles, not precise population tail estimates. Sampled RSS is a lower bound, and a process-lifetime peak is not a per-trial peak. Nested map-query time is already included in rollout time and must not be added again. GPU timing and GPU memory do not apply to this CPU experiment.

**Observed host interruption.** One original provider-cold trial (terrain `loo-surface`, repetition 0) overlapped a confirmed host sleep/resume event. Its saved `perf_counter_ns` interval is 2315.996 s for an actual final target time of 1801 s; the host-event timestamps are 30 min 43 s apart. The overlap was established by mapping the saved counter endpoints using a later paired clock observation, not by original stored UTC endpoints. Artifact completion does not certify uninterrupted timing. We retain the original value, charges and five-trial denominator, without subtracting sleep, excluding the trial or rerunning it. This trial and any affected quantile or normalized mean describe interruption-exposed elapsed time, not isolated uninterrupted prediction latency or evidence of a speedup. The nested terrain-query span shares this limitation. Other trials have not thereby been certified free of interruption. All 165 original timing records are complete, but uninterrupted latency is not certified by that inventory: this is a retained evidence limitation, not an unfinished trial queue. This timing disclosure alone neither recomputes nor independently validates the saved predictive scores.

The same later-clock mapping additionally flags `all-terrain` cold repetition 3 (3000.464 s, actual target 1801 s) in a partial 153-of-165 saved-record census. Host clock-change records also exist, so these mappings are not independent UTC provenance certification. Both flagged subjects require the interruption caveat in affected cost summaries; absence of a flag does not certify uninterrupted execution. Only saved scalar metadata was read, without recomputing scores or loading prediction arrays.

## 11 Completed saved-output replay and current review results

This dated update (9 October 2026) supersedes earlier pending-audit and unfinished-auxiliary descriptions, without rewriting the historical tables. The independent consumer freshly checked all 11368 common-score slots, 58 closed blocks and the complete 11659-item execution inventory against saved outputs. It did not reuse producer-cache numbers for the replay. The original analysis and completed audit were then reused for export; no forecast, fit or repeated audit was started for this review update. This verifies saved-output consistency, not independent participants, physical-clock provenance, numerical-resolution invariance or human acceptance.

Table 4: Current terminal execution and delivered review material. Failed scientific slots are retained, not zero errors.

| Item | Count | Interpretation |
|---|---|---|
| Scientific forecasts | 11020 | 11015 successes, five retained failures |
| Common-score slots | 11368 | 58 blocks; references are additional roles |
| Original auxiliary tasks | 261 | All successful; 38 replays, 58 inertial paths, 75 cold, 15 excluded warmups, 75 warm trials |
| Public CSV files | 16 | All three origin modes; unavailable families retained |
| Figure groups | 16 | Each in PNG and PDF, giving 32 images |
| Primary method contrasts | 21 | 14 practical-equivalence algorithm states, seven inconclusive; no benefit or harm state |

Figure 1 displays every primary method contrast, with negative candidate-minus-control ES favouring the candidate. The equivalence states use the fixed practical band and original gates, not merely a non-significant test. All 21 resolution-invariant verdicts remain inconclusive; the audit does not upgrade them. Both primary terrain families remain wholly unavailable: overall/LOO has 1379 successful rows and one failure out of 1380; LIO has 1148 successful rows and two failures out of 1150. Their 230 baseline rows are shared. We do not estimate effects from a successful intersection or substitute zero for a failed comparison.

All 165 original timing records are now present, including 15 excluded warmups and five trials per cold/warm condition. The 30 condition summaries retain original values and denominators. Host-interrupted elapsed times are not corrected or discarded. The archived runtime plot's *Isolated cold/warm runtime* title is a historical design label, not certification of uninterrupted isolated latency or speedup; the clock and interruption limitations in Section 10 still apply.

The complete anonymous cards, original 16 CSV files, 32 images and 34 bilingual optional table fragments are supplied under `paper/pirc17/final-results-v1/`. They retain all modes, diagnostic witnesses, trial statuses and units. The evidence cards bind analysis `d1d67a3b`, audit `8ff917ad` and export `29cb892f`; full digests and file manifests are in the package. This is a review delivery, not complete final manuscript integration or final scientific/human acceptance.

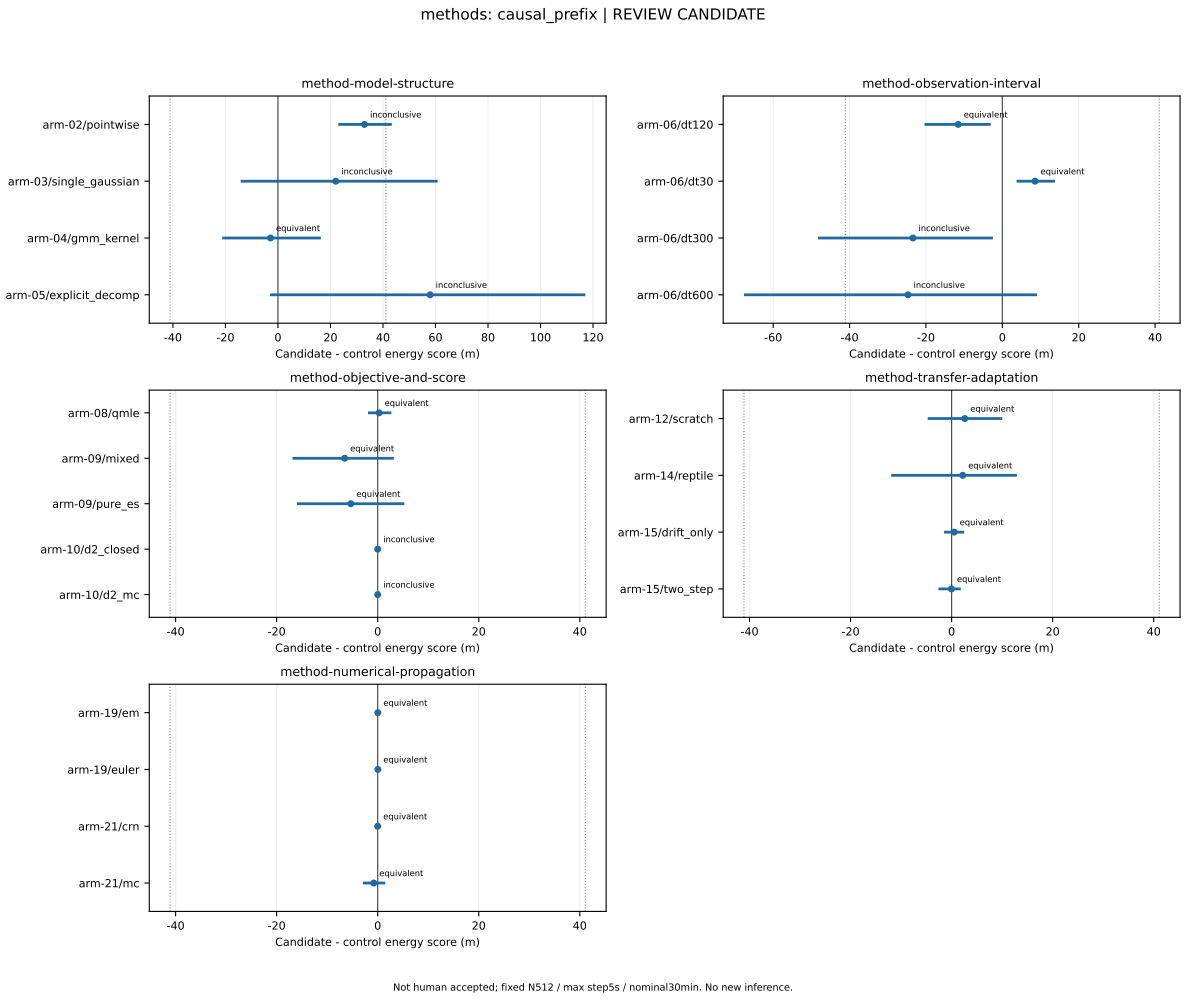



Figure 1: Actual audit-bound primary method comparison plot. Intervals and the fixed practical band are retained from the original analysis. Displayed verdicts are finite-algorithm recorded states, not resolution-invariant qualification or accepted scientific claims.

## 12 Version and evidence binding

The methods in this draft were checked against protocol content identity `2267ab84fa77ff4039234a61e8f44be1c3713317d8ad2d1bb4ced9af0c5b293a` and execution identity `c3e44393f6bc5544aaa21bb273c94eaecb9d0becb4e0590771355c4c0be9b6c2`. The original completed analysis, saved-output audit, public export and cards are bound in `paper/pirc17/final-results-v1/`; later fixed-budget arithmetic and card-link checks are in `paper/pirc17/qualification-v1/`. The unchanged complete anonymous input is now published losslessly in `paper/pirc17/replay-input-v1/`. Isolated standard-library commands there reproduce all six CRPS/endpoint-quantile package files byte-for-byte, not every original paired verdict, figure and PDF. That complete public rebuild path remains to be independently verified; source hashes alone cannot replace it. The executable semantics are in the PIRC-17 modules for direct linear fitting, causal rollout, metric registry, block inference and evidence-card generation. An absence of raw private data from this paper is a publication boundary, not a claim that no independent saved-output audit is needed.

## 13 Release and source-reconciliation details

The following source and bookkeeping descriptions retain the original reported scope. They add no new data qualification or experiment; the main text retains the scientific population, input definitions and limitations.

**Released identities and development assignment.** We checked every one of the 7,618 assignments against this frozen source function and joined every one of the 72,046 sample identities back to its file, hash and split. The file, recording-hash, segment and sample identity intersections are zero for each pair of released splits. Counts and source bindings are in `paper/pirc17/partition-description-v1.json`; that identity reconciliation did not reread raw coordinate, speed or clock values, so it is not a fresh raw-hash, point-alignment, participant or temporal-independence audit.

For a refined segment with $n \geq 4$ points, the single midpoint sample uses history indexes $0, \ldots, \max(1, \lfloor n/2 \rfloor - 1)$, followed immediately by target indexes through $n - 1$. All saved bounds satisfy this rule, and sample identities bind these bounds; each sampled segment has exactly one sample. This excludes history–target index overlap within that sample, not physical-time overlap or approximately duplicated tracks across recordings. The training/adaptation rule is applied to the full released training sample population before task qualification: sort its 5,251 distinct hash blocks by SHA256 of `20260912:` followed by the block ID, assign the first $\text{round}(0.8 \times 5251) = 4201$ blocks to training, and the remaining 1,050 to adaptation. They contain 39,272 and 10,573 windows, respectively. The selected 328 training windows occupy 259 blocks, 76 adaptation windows occupy 62 blocks, and 81 validation windows occupy 62 blocks. All 16 saved method fits agree with these original roles and window identities. This does not replace the much smaller actual fitting counts with the full release counts.

**Evaluation-use history.** The release report records 74 exact recording-hash groups crossing the *legacy directory* splits; that is a historical reported audit, not the current zero-intersection assignment above. The bound usage history also records prior exposure to the 12,370-window released final-evaluation cohort. The separately used historical GeoLife evaluation had 17,038 windows and is not a cohort in this protocol. The owner elected on 26 September 2026 to use the already downloaded data with separated training/evaluation, rather than create a new untouched holdout.

We retain that scope, but do not describe this evaluation as first-use, design-blind validation. A zero-read declaration for protocol preparation does not erase historical use or its limits on claims of independent confirmation.

**Saved-speed construction.** The source construction script converts input km/h to m/s and repeats the last interval speed at the parent segment's endpoint. That padding and upstream processing are retained; we do not regenerate or repair speeds. Its separate $v_x, v_y$ columns use centered finite differences of local positions with copied endpoint velocities, so their norm is not the saved scalar summarized here. We do not estimate speed using the release clock. All 12,370 original windows have complete finite nonnegative saved scalars; no window is removed and no failed-stage distribution is replaced by a successful-subset estimate.

**Clock footers and reported anomalies.** Table scientific supplement separates the saved and algorithmic clocks. The release casts `t` to `datetime64[ns]` and integer ticks; the formal input loader checks exact equality with the qualified source-point ticks. A dtype or equality check is not a timezone/provenance certificate. In a fixed first-three-file footer inspection of the original condition manifest, all three hash-matched files have `timestamp[ns]` with no timezone annotation (951, 550 and 794 rows). This limited inspection reads no point values and does not describe all 7,618 files. A naive timestamp may have been stored under a UTC convention, but the footer does not establish that convention. No new timezone correction is inferred or applied.

The original release report records 37,957 duplicate-time intervals in 1,387 segments, 166 backward-time cuts and 10,052 gaps greater than 60 seconds causing cuts. It reports 19,647,139 source condition points, 14,738,300 aligned points and 4,908,839 points not aligned. These are reported release-reconciliation totals, not fresh pointwise recomputations, clock error rates or anomaly counts for the selected 46 targets. R1 cleaning, time refinement and qualification have different denominators; their counts must not be pooled as one error rate. The legacy observation adapter keeps the first exact-time duplicate and balances its history/target sides by selecting existing endpoints. In contrast, the formal visible-prefix loader requires strictly increasing retained times and does not silently deduplicate a new prediction input. Metadata, source bindings and the three footer receipts are in `paper/pirc17/clock-description-v1.json`.

**Case maps and model asset accounting.** The private case backgrounds use WorldCover 2021 v200[16], SRTM-labelled Skadi elevation[20], Overture transportation release 2026-08-19.0 (including OpenStreetMap and TomTom sources)[23, 17], and HydroRIVERS v1[18, 19]. These fixed map vintages are not demonstrated to be contemporaneous with each source trajectory. The route-free acknowledgement record `paper/pirc17/map-source-acknowledgements-v1.json` distinguishes WorldCover's CC BY 4.0, transportation's ODbL, the SRTM receipt declarations, and HydroRIVERS' product-specific agreement; it does not assign one common licence to all layers. HydroRIVERS' prescribed notices and source citations still require publication review. Map availability and credits do not establish permission to redistribute private trajectories. Original-route acquisition permission, privacy review and case publication authorization remain unverified; this is a disclosure, not an assertion that publication has been cleared.

The model's retained map catalog is distinct from the three illustrated backgrounds. Table scientific supplement describes the five hash-bound acquisition receipts and additional raster parents from the frozen feature snapshot. Counts are admitted files, not assets actually visited, covered trajectories or successful queries. The same 1283 snapshot parents overlap 1232 receipt assets; only

51 add files to the 1354-file receipt catalog. The resulting 1405-file catalog does not guarantee support at predicted positions.

**Product references and upstream provenance.** The model's elevation channel labelled `srtm` uses the Mapzen AWS Skadi distribution[20]; the separate admitted Copernicus channel is GLO-30 Public, 2021 release[22]. WorldCover v200[16], Overture release 2026-08-19.0[23], and HydroRIVERS v1.0[18, 19] identify the other model-input products, not just case backgrounds. The Skadi format documentation describes a distribution using SRTM and other open elevation sources. The receipt's GL1 label and 2000 interval therefore do not certify pure NASA SRTM provenance or acquisition year for every retained tile. We retain the frozen asset hashes and receipt labels; the cited documentation revision is not a newly assigned data release. Provider attribution documents[21, 17, 19] are distinct from product and methodological references. Their consultation does not clear tile-specific obligations or private-route publication. Documentation access dates are not acquisition dates or immutable data versions; missing per-tile upstream identities remain a limitation.

## 14 Scoring and execution capture times

The scoring snapshot captured on 6 October 2026 at 00:54:18.683579 UTC contains 10384 successful scientific score rows, five failure rows and 631 missing rows from the registered 11020 forecasts. These are captured scoring dispositions, not a real-time forecast counter. The entire primary method grid is complete: 28 configurations, 46 recording-hash blocks and five forecast RNG seeds give 6440 score rows. Each configuration has 230 rows, but the recording-block count is 46. No forecast, refit or particle scoring was performed for this snapshot. Its precise capture time and input hashes are retained in `paper/pirc17/preliminary-method-statistics-v1.json`. The execution census in Section 15 was captured later, at 06:53:13.846553 UTC; its success count is 130 higher. These are different dated snapshots, not two simultaneous counters. Their item-level difference has not been independently audited here.

Table 5 reconciles the two saved summaries on the identical 114 configuration–origin-mode cells. All 22 cells with changed counts belong to the terrain matrix: their net success increases are 57 for causal prefixes, 8 for known velocity and 65 for point-only origins. In each cell the success increase equals the decrease in missing counts, and the failed count is unchanged. All 84 method cells are unchanged. Thus the 130-count difference does not add observations to the 6440-row primary method result tables or change their pinned input version. The corresponding abstract and method findings retain the earlier scoring snapshot; the completeness tables retain the later execution census.

The route-free `paper/pirc17/capture-count-reconciliation-v1.json` retains every cell's two count triples, exact capture times and input file hashes. Neither original summary retains the per-item work-ID sets. Consequently a net increase of 130 is not a verified set of 130 new item identities: membership replacement within a cell cannot be ruled out by these counts. This count reconciliation neither replaces the original saved-output audit nor establishes payload integrity, independence or scientific qualification. No source snapshot or statistical result is overwritten, and no prediction, score or resampling is added.

Table 5: Reconciliation of saved success counts, not a reconstruction of individual forecast membership. Earlier means the 00:54 scoring summary; later means the 06:53 execution census on 6 October 2026.

| Matrix | Origin mode | Earlier | Later | Net change |
|---|---|---|---|---|
| Methods | Causal prefix | 6440 | 6440 | 0 |
| Methods | Known velocity | 838 | 838 | 0 |
| Methods | Point only | 840 | 840 | 0 |
| Terrain | Causal prefix | 1974 | 2031 | 57 |
| Terrain | Known velocity | 292 | 300 | 8 |
| Terrain | Point only | 0 | 65 | 65 |
| Total | | 10384 | 10514 | 130 |

## 15 Captured forecast and family completeness

Table 6 is a captured execution census at 6 October 2026, 06:53 UTC, not a new accuracy analysis. It reads only the original workload, prediction and score-cache indexes, including exact saved-source bindings. It neither loads prediction arrays nor re-estimates scores. All 11020 planned scientific forecasts retain their original configuration, origin-mode and five-seed denominator. At capture, 10514 were successful, 501 still runnable and five failed; their score indexes had the same counts, with no invalid bindings detected. An indexed success is not independent validation of the score payload or scientific qualification.

Table 6: Unique planned forecasts by matrix and origin mode at the captured time. These six rows sum to 11020; origin modes reuse blocks, not new people.

| Matrix | Origin mode | Planned | Successful | Runnable | Failed |
|---|---|---|---|---|---|
| Methods | Causal prefix | 6440 | 6440 | 0 | 0 |
| Methods | Known velocity | 840 | 838 | 0 | 2 |
| Methods | Point only | 840 | 840 | 0 | 0 |
| Terrain | Causal prefix | 2300 | 2031 | 266 | 3 |
| Terrain | Known velocity | 300 | 300 | 0 | 0 |
| Terrain | Point only | 300 | 65 | 235 | 0 |
| Total | | 11020 | 10514 | 501 | 5 |

The method results in Section scientific supplement concern the complete $28 \times 46 \times 5 = 6440$ primary grid, not a mean over all matrices and modes. Known-velocity and point-only cases each reuse the first six primary blocks and are bounded descriptive initialization diagnostics. The known-velocity failures are in `single_gaussian` and `dt30`; the primary terrain failures are in `all-terrain`, `lio-road` and `lio-river`. They are retained, not retried automatically or omitted from denominators.

Table 7: Required primary score-index rows in each registered comparison family at the same capture. Complete means index-complete only. Missing or failed rows prevent a complete paired comparison; families overlap.

| Primary family | Required | Success | Missing | Failed | Disposition |
|---|---|---|---|---|---|
| Model structure | 1150 | 1150 | 0 | 0 | Complete index |
| Observation interval | 1150 | 1150 | 0 | 0 | Complete index |
| Objective and score | 1380 | 1380 | 0 | 0 | Complete index |
| Training and adaptation | 1150 | 1150 | 0 | 0 | Complete index |
| Numerical propagation | 1380 | 1380 | 0 | 0 | Complete index |
| Terrain all/LOO | 1380 | 1219 | 160 | 1 | Unavailable |
| Terrain LIO | 1150 | 1017 | 131 | 2 | Unavailable |

Table 7 counts the unique required configuration rows within each family, not one new set of observations per contrast. Shared controls appear in more than one family, so its rows cannot be summed into an experiment total. In particular, the terrain families share 230 baseline rows; their 160 and 131 missing scores include 25 shared baseline rows and therefore represent 266 unique primary missing forecasts, not 291. Index completion does not waive the independent saved-output audit, mechanism evidence, uncertainty checks or final acceptance. No effect estimate is obtained from a failed family's successful intersection.

The 501 remaining scientific forecasts are separate from the 261 scheduled auxiliaries: 58 inertial paths, 38 replays, 75 cold timings, 15 excluded warm-ups and 75 warm timings. None of those auxiliaries had completed at capture; 290 same-grid references had already completed. Warm-up measurements are excluded from the measured runtime sample. Cost conclusions must use the original timing protocol, not scientific prediction counts or this census.

## 16 Detailed CPU environment records

Recorded threads are PyTorch 6/inter-op 12, OMP/MKL 6 and OpenBLAS/Numba 12. Library thread counts are not prediction-worker counts. The machine descriptor identifies Intel family/model/stepping, not a verified marketing CPU model. The public reader projection binds the original environment digest and removes workstation/DLL paths.

Each isolated CPU trial carries the original digest of a hardware manifest: platform and processor reports, logical and physical CPU counts, total RAM, Python, NumPy, PyTorch and psutil versions, and versions of available DuckDB, Numba, Rasterio and PyArrow packages. It also records PyTorch thread counters, Numba's thread count if loaded, and configured OMP, MKL, OpenBLAS and Numba thread-environment variables. An unset variable does not imply a single thread. The public hardware entry is keyed by the original full-manifest digest and omits loaded native-threadpool records that can contain local DLL paths. Final hardware values must come from the audited timing records, not the machine that compiles this draft. The timing protocol's "warm OS cache" wording is not a measured cache-state guarantee: trials do not flush, control or record OS cache occupancy. The original timing receipts and their definitions are retained, not relabelled or remeasured.

## 17 Execution records and reconstruction

During continuation, the two workers process disjoint original task identities. Each owns its mutable map-query and history state; one dispatcher writes atomic completion metadata. Thirty-minute monitoring records overall success/failure/remaining counts and separate worker increments. These are execution observations, not model-quality scores or additional experiments.

The closed saved-score inventory and the original scientific-task metadata agree on the terminal counts in Table 8. The 8120 method forecasts and 2900 terrain forecasts total 11020 scientific workloads. The common-score grid additionally includes 290 registered same-grid references and 58 deterministic inertial paths; it therefore has $58\{5(28 + 10 + 1) + 1\} = 11368$ slots, not 11368 independent observations. There are 46 primary and 12 support blocks. A failed forecast retains a failed score slot, not a zero-valued error. Counts do not repair the four unavailable comparison families or establish that every score agrees with the raw output.

Common scoring has closed all 11368 rows and 58 blocks; all 261 original auxiliary tasks and the independent saved-output replay have completed. The once-only original analysis and audit are retained unchanged. Public export and evidence-card construction have completed without new forecasts, fits or a repeated audit. Section 11 identifies the current complete review package. The public consumer produces sixteen tables and sixteen figures in PNG and PDF, retaining the original estimates, intervals, missingness and units. Registered aggregate exports omit private trajectories, targets, coordinates, model weights and local machine paths. Their content identities and rendering environment permit an independent reconstruction without reopening private data.

Reproduction has three distinct scopes. Rebuilding the manuscript compiles the declared English and Chinese sources and resolves their references; it does not repeat forecasts. Regenerating

Table 8: Terminal saved-score inventory, inspected 8 October 2026. Expected counts retain all registered slots; success/failure are saved dispositions, not scientific qualification. References are outside the 11020 scientific-forecast denominator.

| Role | Expected | Successful | Failed |
|---|---|---|---|
| Method forecasts | 8120 | 8118 | 2 |
| Terrain forecasts | 2900 | 2897 | 3 |
| Scientific subtotal | 11020 | 11015 | 5 |
| Same-grid references | 290 | 290 | 0 |
| Inertial paths | 58 | 58 | 0 |
| Common-score total | 11368 | 11363 | 5 |

public tables and figures consumes the hash-bound evidence cards, copies the original numerical estimates and retains explicit unavailable cells, without new bootstrap draws or tests. Scientific reproduction additionally checks saved forecast outputs and their source bindings through the independent audit. A successful PDF build cannot replace that audit or authorize an empirical claim.

The final review record will identify the immutable public export and cards, their input manifest, protocol and execution, renderer versions and exact manuscript revision. Figure byte equality is limited to the recorded library and font environment; arbitrary TeX installations need not produce identical PDF bytes. Historical manuscripts remain separately named, and their existing release workflow is not evidence that this PIRC-17 draft has been approved or published. Final authorship and acceptance remain explicit human decisions.

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